Continuous-Time Portfolio Allocation

Merton's Close-Form Solution, Simulation, and Empirical ETF Backtesting

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1 Introduction

We study a continuous-time portfolio allocation problem inspired by Merton's Portfolio Problem (1969). We assume a continuous-time market model in which risky assets follow Geometric Brownian Motion, a risk-free asset exists, and portfolios are continuously rebalanced. Within the classical Merton framework, we derive the wealth dynamics and obtain the closed-form solution for the optimal allocation between risky assets and the risk-free asset. The project focuses on obtaining this optimal portfolio allocation by maximizing expected utility of terminal wealth under constant relative risk aversion (CRRA) preferences.

We validate the analytical solution through Monte Carlo simulations and extend the framework to a multi-asset setting. Using empirical ETF data from historical observations, we evaluate the out-of-sample performance of the implied Merton allocation relative to an equal-weight benchmark. The results illustrate both the theoretical structure of optimal allocation and the practical importance of estimation risk and leverage constraints. Ongoing work focuses on improving covariance estimation through shrinkage techniques, incorporating long-only constraints, and extending the framework toward dynamic consumption–investment problems via the Hamilton–Jacobi–Bellman (HJB) equation.

2 Continuous-Time Market Model and Wealth Dynamics

Risk-Free Asset

The risk-free asset evolves deterministically as

$$dB_t = rB_t\,dt, \tag{1}$$

$$B_t = B_0 e^{rt}, \tag{2}$$

where r is the constant continuously-compounded risk-free rate.

Risky Asset

The risky asset price S_t follows a Geometric Brownian Motion (GBM):

$$dS_t = \mu S_t\,dt + \sigma S_t\,dZ_t, \tag{3}$$

where:

- μ is the constant drift (expected return),
- $\sigma > 0$ is volatility,
- $\{Z_t\}_{t\geq 0}$ is a standard Brownian motion.

In the explicit solution, S_t is lognormally distributed.

$$S_t = S_0 \exp\left(\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma Z_t\right), \tag{4}$$

2.1 Wealth Dynamics

Let W_t denote total investor wealth at time t. The investor allocates a constant fraction w of wealth to the risky asset and $1 - w$ to the risk-free asset.

Under a self-financing strategy, wealth evolves as

$$dW_t = (1-w)W_t r\,dt + wW_t(\mu dt + \sigma dZ_t). \tag{5}$$

$$dW_t = W_t\left[r + w(\mu - r)\right]dt + w\sigma W_t dZ_t. \tag{6}$$

From the wealth dynamics above, we can rewrite:

$$dW_t = aW_t dt + bW_t dZ_t, \tag{7}$$

where

$$a = r + w(\mu - r), \qquad b = w\sigma, \qquad \{Z_t\}\text{: std.Brownian motion}$$

This is a linear stochastic differential equation identical in structure to GBM. Here W_t is an Itô's process.

Application of Itô's Lemma

To obtain the explicit solution, we apply Itô's lemma to the function $f(W_t) = \log W_t$.
For a twice-differentiable function f, Itô's lemma gives

$$df(W_t) = f'(W_t)dW_t + \frac{1}{2}f''(W_t)(dW_t)^2.$$

Since

$$f'(x) = \frac{1}{x}, \qquad f''(x) = -\frac{1}{x^2},$$

Also, noting that $(dZ_t)^2 = dt$, we obtain

$$d(\log W_t) = \frac{1}{W_t}(aW_t dt + bW_t dZ_t) - \frac{1}{2}\frac{1}{W_t^2}b^2 W_t^2 dt \tag{8}$$

$$= \left(a - \frac{1}{2}b^2\right)dt + bdZ_t. \tag{9}$$

Distribution of terminal wealth

To arrive at terminal wealth, say at time T, we integrate from 0 to T yields

$$\log W_T = \log W_0 + \left(a - \frac{1}{2}b^2\right)T + bZ_T. \tag{10}$$

Exponentiating both sides gives the explicit solution

$$W_T = W_0 \exp\left(\left(a - \frac{1}{2}b^2\right)T + bZ_T\right). \tag{11}$$

Substituting a and b:

$$W_T = W_0 \exp\left(\left[r + w(\mu - r) - \frac{1}{2}w^2\sigma^2\right]T + w\sigma Z_T\right). \tag{12}$$

Since $Z_T \sim \mathcal{N}(0, T)$, it can be clearly seen that W_T is log-normal:

$$\log W_T \sim \mathcal{N}(m, v),$$

where

$$m = \log W_0 + \left[(1-w)r + w\mu - \frac{1}{2}w^2\sigma^2\right] T, \tag{13}$$

$$v = w^2\sigma^2 T. \tag{14}$$

Therefore, terminal wealth W_T is lognormally distributed.

The term $-\frac{1}{2}w^2\sigma^2$ arises from the quadratic variation of Brownian motion and reflects the impact of volatility on long-run geometric growth.

3 Optimal Allocation

3.1 Expected Utility Maximization - CRRA Risk Aversion

To evaluate optimal portfolio allocation, we use an expected utility-based framework, Constant Relative Risk Aversion (CRRA), to maximize expected utility over terminal wealth. Note that, if we would have maximized expected wealth $E[W_T]$, optimal solution would just be infinite leverage (if $\mu > r$). CRRA utility function:

$$U(W_T) = \frac{W_T^{1-\gamma}}{1-\gamma}, \qquad \gamma > 0,\ \gamma \neq 1. \tag{15}$$

CRRA utility functions are concave and provides a stable, and tractable close-form solution. The implication is that the proportion of wealth invested in risky assets remains constant even as wealth changes.

The parameter γ denotes the coefficient of relative risk aversion, and measures curvature. Higher values of γ correspond to more curvature and greater aversion to risk. Low γ implies flatter utility, and more risk tolerance, i.e., willing to accept volatility in exchange for potential gains.

To maximize the expected utility:

$$\max_w \ \mathbb{E}\left[W_T^{1-\gamma}\right]. \tag{16}$$

Note that the multiplicative constant $\frac{1}{1-\gamma}$ does not affect the optimizer.

Build plots for expected utility.

Computation of Expected Utility

From the previous section,

$$\log W_T \sim \mathcal{N}(m, v).$$

Observe that

$$W_T^{1-\gamma} = \exp\left((1-\gamma)\log W_T\right).$$

If $X \sim \mathcal{N}(m, v)$, the moment-generating property of the normal distribution gives

$$\mathbb{E}[e^{aX}] = e^{am+\frac{1}{2}a^2v}.$$

Applying this with $a = 1-\gamma$ yields

$$\mathbb{E}[W_T^{1-\gamma}] = W_0^{1-\gamma} \exp\left((1-\gamma)\left[r + w(\mu - r)\right]T - \frac{1}{2}(1-\gamma)\gamma w^2\sigma^2 T\right). \tag{17}$$

The expected utility of a utility function represents the anticipated satisfaction an individual expects to derive from an outcome characterized by uncertainty, meaning the individual is risk-averse.

Closed-form Solution

Since the exponential function is monotonic, maximizing expected utility is equivalent to maximizing the exponent, which is a function of w:

$$f(w) = (1-\gamma)w(\mu - r)T - \frac{1}{2}(1-\gamma)\gamma w^2\sigma^2 T. \tag{18}$$

Taking the first derivative,

$$f'(w) = (1-\gamma)(\mu - r)T - (1-\gamma)\gamma w\sigma^2 T. \tag{19}$$

Setting $f'(w) = 0$ gives the optimal allocation

$$w^* = \frac{\mu - r}{\gamma\sigma^2} = \frac{\text{Risk Premium}}{\text{Risk Aversion} \times \text{Variance}} \tag{20}$$

The second derivative is negative for $\gamma > 0$, confirming a maximum.

Higher expected excess return increases risky exposure, while higher volatility or greater risk aversion reduces it.

Note that the optimal allocation is constant and does not depend on wealth or time. This tractability arises from the lognormal structure of wealth and CRRA preferences. This is exactly what defines the constant allocation and continuous-time rebalancing.

4 Multi-Asset Extension

We now extend the framework to n risky assets (say $\mathbf{S}_t = (S_t^1, \ldots, S_t^n)^\top$). Let:

- $\boldsymbol{\mu} \in \mathbb{R}^n$ denote the vector of expected returns,
- $\Sigma \in \mathbb{R}^{n\times n}$ denote the positive-definite covariance matrix,
- r denote the constant risk-free rate.

Let $\mathbf{w} \in \mathbb{R}^n$ denote the constant vector of portfolio weights allocated to risky assets. The remaining wealth $1 - \mathbf{1}^\top\mathbf{w}$ is invested in the risk-free asset.

The wealth dynamics become

$$\frac{dW_t}{W_t} = \left[r + \boldsymbol{w}^\top(\boldsymbol{\mu} - r\mathbf{1})\right]dt + \boldsymbol{w}^\top\Sigma^{1/2}d\boldsymbol{Z}_t, \tag{21}$$

where $\boldsymbol{Z}_t$ is an n-dimensional standard Brownian motion.

The terminal wealth remains lognormal. Over terminal horizon $[0, T]$, we get:

$$W_T = W_0 \exp\left(\left[r + \mathbf{w}^\top(\boldsymbol{\mu} - r\mathbf{1}) - \frac{1}{2}\mathbf{w}^\top\Sigma\mathbf{w}\right]T + \mathbf{w}^\top\Sigma^{1/2}\mathbf{Z}_T\right). \tag{22}$$

Optimal Allocation under CRRA Preferences : Following the same steps as in the single-asset case, terminal wealth remains lognormally distributed, and expected utility depends on the mean and variance of log-wealth. Maximizing expected CRRA utility again, yields the closed-form solution:

$$\mathbf{w}^* = \frac{1}{\gamma}\Sigma^{-1}(\boldsymbol{\mu} - r\mathbf{1}). \tag{23}$$

The optimal portfolio is proportional risk premium and inversely proportional to covariance-matrix. Assets with higher excess return receive greater weight, while assets contributing more covariance risk receive lower weight. Multi-Asset Weight Looks Like Mean-Variance

5 Monte Carlo Validation

5.1 Single-Asset Market Model

To validate the analytical solution, we conduct a Monte Carlo simulation of the wealth process under continuous-time rebalancing.

Model Parameters

The simulation uses the following parameter values:

- Expected return: $\mu = 0.08$
- Risk-free rate: $r = 0.02$
- Volatility: $\sigma = 0.20$
- Risk aversion: $\gamma = 3$
- Horizon: $T = 1$
- Initial wealth: $W_0 = 1$

The analytical optimal allocation (from closed-form) is:

$$w^* = \frac{\mu - r}{\gamma\sigma^2} = 0.50$$

Monte-Carlo sampling to validate Expected Utility

We simulate the wealth process using the exact discretization of the lognormal dynamics:

$$W_{t+\Delta t} = W_t \exp\left(\left[r + w(\mu - r) - \frac{1}{2}w^2\sigma^2\right]\Delta t + w\sigma\sqrt{\Delta t}\, Z\right),$$

where $Z \sim \mathcal{N}(0, 1)$ independently across time steps and paths.

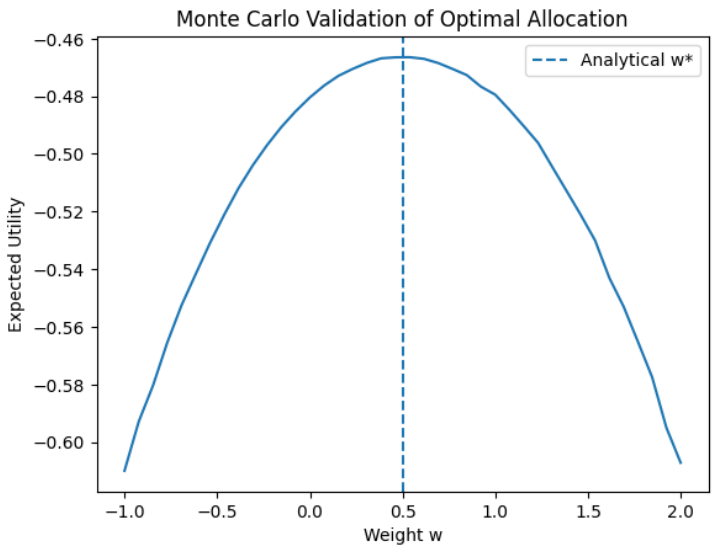

Figure 1: The numerical optimum aligns closely with the theoretical allocation

For each candidate allocation w in a predefined grid, we:

1. Simulate N independent wealth paths up to time T. We use 252 time steps (daily frequency), and 100,000 independent paths.
2. Compute terminal wealth $W_T^{(i)}$ for each path $i = 1, \ldots, N$.
3. Evaluate CRRA utility:
$$U\big(W_T^{(i)}\big) = \frac{\big(W_T^{(i)}\big)^{1-\gamma}}{1-\gamma}.$$
4. Estimate expected utility via sample average:
$$\widehat{\mathbb{E}}\big[U(W_T)\big] = \frac{1}{N}\sum_{i=1}^{N} U\big(W_T^{(i)}\big).$$

Results

The simulated expected utility is plotted as a function of w and exhibits a concave shape, with the maximum occurring near the analytical solution (0.5).

5.2 Sensitivity Analysis : Role of Risk Aversion parameter γ

The optimal allocation (w) derived earlier is

$$w^*(\gamma) = \frac{\mu - r}{\gamma\sigma^2}. \tag{24}$$

Differentiating with respect to γ yields

$$\frac{\partial w^*}{\partial \gamma} = -\frac{\mu - r}{\gamma^2\sigma^2} < 0. \tag{25}$$

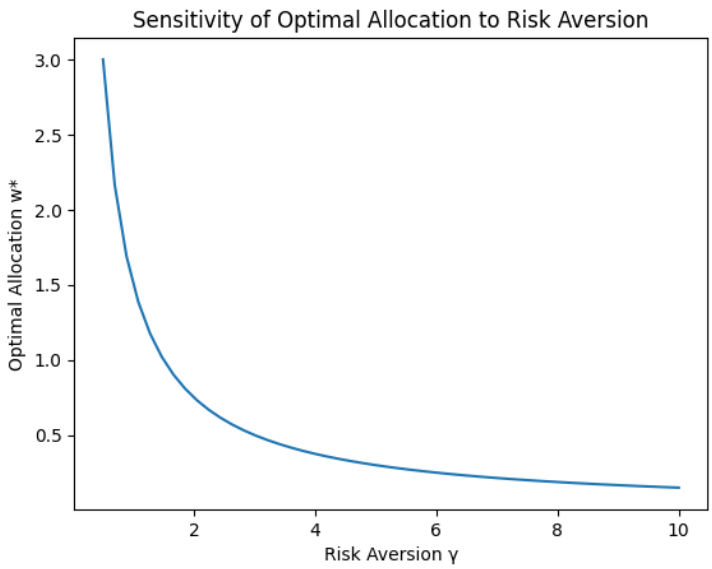

Figure 2: Optimal risky asset allocation (w) is strictly decreasing with γ

Thus, the optimal exposure to the risky asset is strictly decreasing in the coefficient of relative risk aversion (γ). Higher risk aversion leads to proportionally lower risky allocation.

In the limit, $\gamma \to \infty \quad \Rightarrow \quad w^* \to 0$, implying full investment in the risk-free asset.

Economic Interpretation

Investors with lower γ tolerate greater volatility and therefore allocate more aggressively to the risky asset, potentially employing leverage when $w^* > 1$ (borrowing from risk-free asset).

Conversely, highly risk-averse investors (large γ) shrink exposure toward zero.

Connection to the Kelly Criterion

When $\gamma = 1$, CRRA utility reduces to logarithmic utility:

$$U(W) = \log W. \tag{26}$$

In this case, the optimal allocation becomes

$$w^* = \frac{\mu - r}{\sigma^2}, \tag{27}$$

which corresponds to the Kelly growth-optimal strategy.

This allocation maximizes expected log wealth and therefore maximizes long-run capital growth. The Kelly criterion plays an important role in leveraged portfolio construction and position sizing in quantitative investment strategies.

5.3 Multi-Asset Market Model

We extend the validation to a market with n risky assets whose returns follow correlated Geometric Brownian Motions. With constant rebalancing to maintain weights w^*, $\log W_T$ is normally distributed and terminal wealth is lognormal.

The wealth solution over horizon T, as calculated before, is

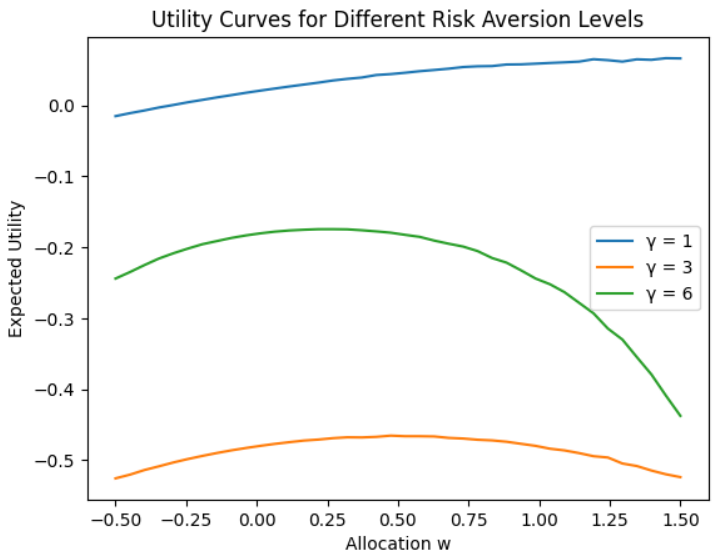

Figure 3:

$$W_T = W_0 \exp\left(\left[r + w^{*\top}(\mu - r\mathbf{1}) - \frac{1}{2}w^{*\top}\Sigma w^*\right]T + w^{*\top}Z_T\right). \tag{28}$$

Under CRRA preferences, the optimal allocation generalizes to

$$w^* = \frac{1}{\gamma}\Sigma^{-1}(\mu - r\mathbf{1}), \tag{29}$$

where γ denotes relative risk aversion and r the risk-free rate.
This allocation:

- Rewards high expected excess return,
- Penalizes high variance,
- Adjusts for correlation across assets,
- Scales inversely with risk aversion.

Simulation Procedure
To validate the analytical solution via Monte Carlo simulation:

1. We compute w^* from the closed-form expression.
2. We then generate correlated Brownian increments using Cholesky decomposition $\Sigma = LL^\top$.
3. We simulate wealth paths under continuous rebalancing, and compute terminal wealth W_T for each path.
4. Finally, we estimate expected utility and validate.

$$\widehat{\mathbb{E}}[U(W_T)] = \frac{1}{N}\sum_{i=1}^{N}\frac{(W_T^{(i)})^{1-\gamma}}{1-\gamma}. \tag{30}$$

Result: The simulated terminal wealth distribution matches the theoretical lognormal structure, and estimated expected utility aligns with the analytical solution.

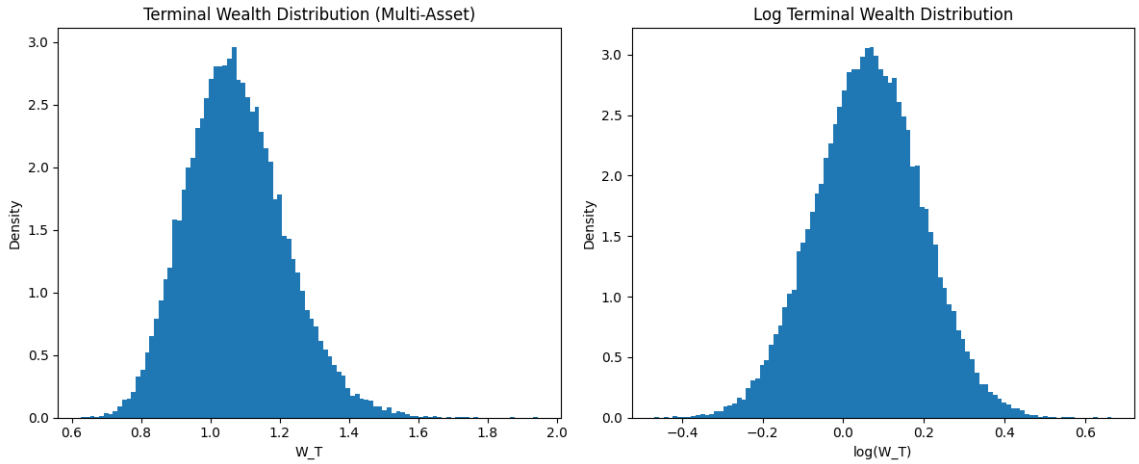

Figure 4:

6 Empirical Multi-Asset Implementation and Backtesting

6.1 ETF Data Description

To evaluate the practical performance of the utility-based allocation, we implement the model using historical ETF data for a few diversified asset classes.

The following liquid exchange-traded funds (ETFs) are used over the market period 2015-2025:

- **Equities:** SPY (US broad market), QQQ (US technology), EFA (Developed international), EEM (Emerging markets)
- **Fixed Income:** TLT (Long-term bonds), IEF (Intermediate bonds)
- **Alternatives:** GLD (Gold), VNQ (REITs)

Daily adjusted closing prices are obtained via the `yfinance` Python API. Adjusted prices are used to account for dividends and corporate actions.

Daily log returns are computed as:

$$r_{t,i} = \log\left(\frac{P_{t,i}}{P_{t-1,i}}\right), \tag{31}$$

where $P_{t,i}$ denotes the adjusted price of asset i at time t.

6.2 Out-of-Sample Backtesting

To avoid look-ahead bias, the dataset is divided into:

- Training period: first 70% of observations (2015-2021)
- Test period: remaining 30% (2022-2025)

Parameter estimation is performed exclusively on the training set, while performance evaluation is conducted on the test set. This constitutes our out-of-sample backtest.

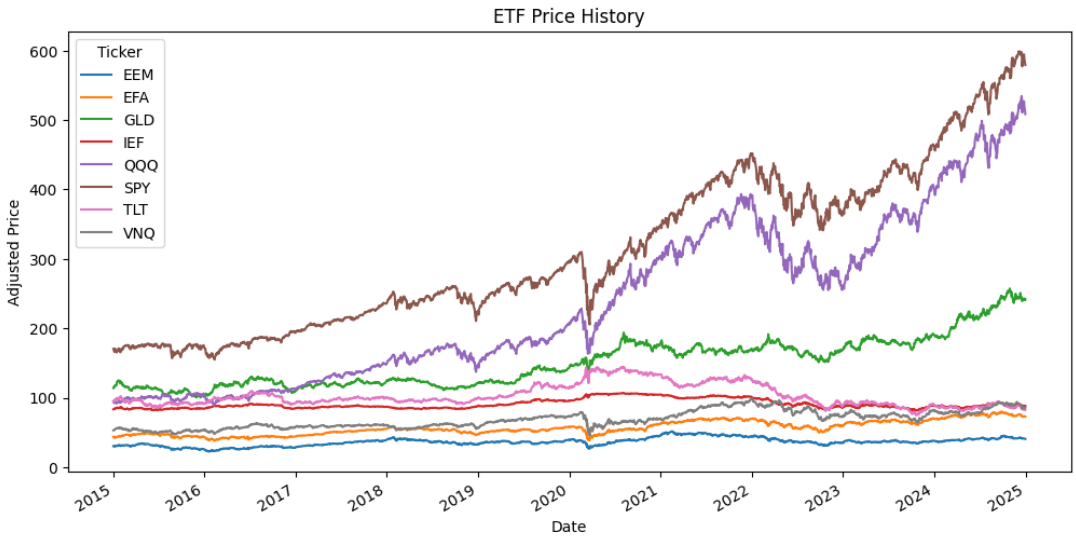

Figure 5:

Parameter Estimation : Mean and Covariance Matrix

Let R_{train} denote the matrix of daily returns in the training sample. We estimate the mean returns vector and covariance matrix from R_{train}.
The annualized mean return vector is estimated as:

$$\hat{\mu} = 252 \cdot \text{mean}(R_{\text{train}}), \tag{32}$$

and the annualized covariance matrix as:

$$\hat{\Sigma} = 252 \cdot \text{cov}(R_{\text{train}}). \tag{33}$$

These empirical estimates serve as inputs to the multi-asset Merton allocation formula.

Backtesting

Using the estimated optimal weight w^*, portfolio returns in the test period are computed as:

$$r_{p,t} = {w^*}^\top r_t, \tag{34}$$

where r_t is the vector of observed test-period returns.
Portfolio wealth evolves according to:

$$W_{t+1} = W_t \left(1 + r_{p,t}\right). \tag{35}$$

This procedure evaluates how the theoretically optimal allocation performs on unseen data.

Performance Metrics

Apart from the terminal wealth comparison, we also compute the following performance statistics are computed over the test period:

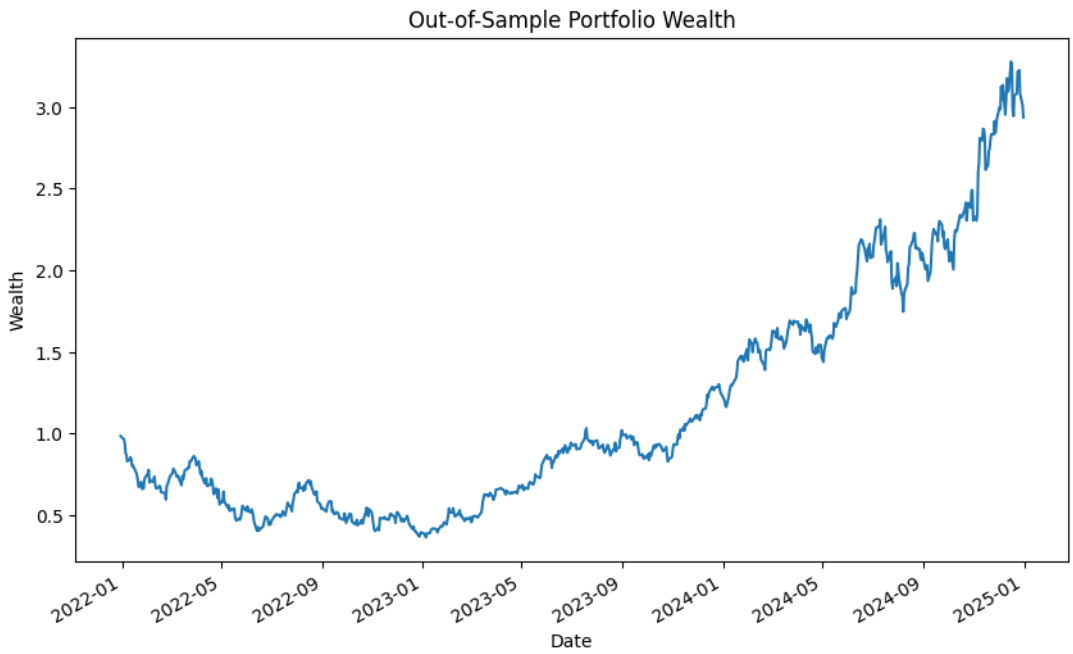

Figure 6: Out-of-sample wealth(log-scale) evolution for the three portfolio strategies.

- Annualized return:

$$\hat{\mu}_p = 252 \cdot \text{mean}(r_{p,t}), \tag{36}$$

- Annualized volatility:

$$\hat{\sigma}_p = \sqrt{252} \cdot \text{std}(r_{p,t}), \tag{37}$$

- Sharpe ratio:

$$\text{Sharpe} = \frac{\hat{\mu}_p - r}{\hat{\sigma}_p}, \tag{38}$$

- Maximum drawdown:

$$\text{MDD} = \min_t \left(\frac{W_t}{\max_{s \le t} W_s} - 1 \right). \tag{39}$$

This empirical implementation bridges continuous-time portfolio theory and practical asset management. While the continuous-time model assumes constant investment opportunities, real-world backtesting reveals the impact of parameter uncertainty, market dynamics, and leverege management on realized performance. We also see the result of optimal weight leading to leverage in risk-free in our Merton's implementation.

6.3 Empirical Backtest Results

Out-of-Sample Performance Summary

Using ETF data from 2015–2025, parameters were estimated on the first 70% of observations and portfolio performance was evaluated on the remaining 30% (out-of-sample).

In addition to the theoretical allocation implied by the continuous-time model, we compare two alternative portfolio constructions:

- Unconstrained Merton allocation (direct implementation of the optimal weights)
- Normalized Merton allocation (weights scaled to sum to one to remove leverage)

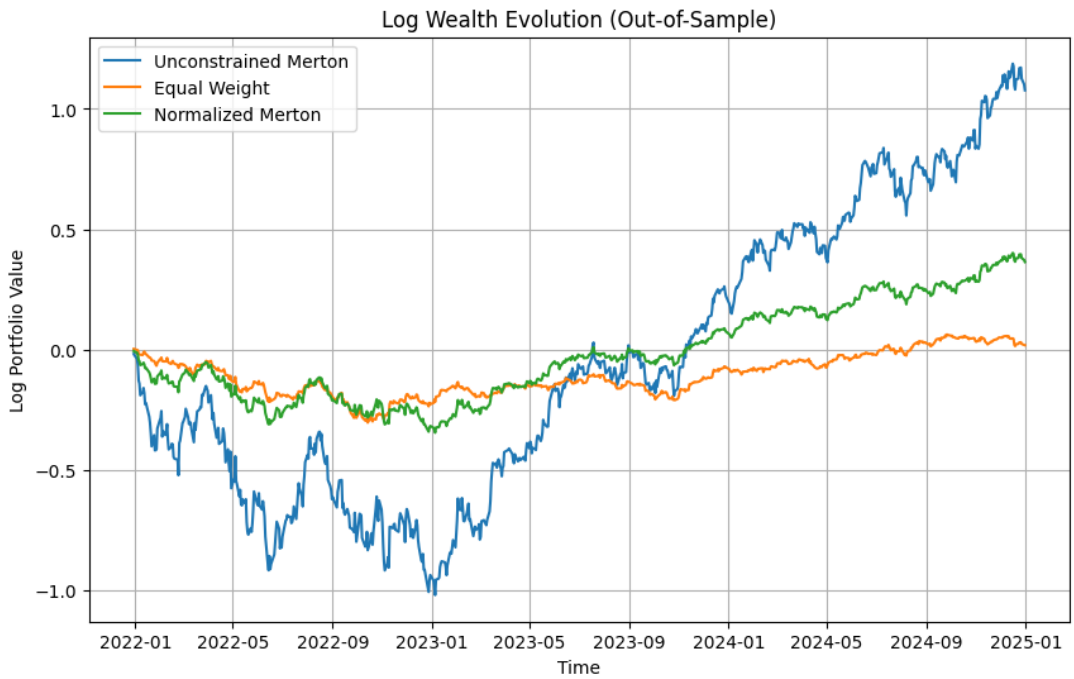

Figure 7: Wealth evolution comparison between portfolio strategies.

- Equal-weight portfolio (benchmark)

The resulting performance statistics are summarized below:

Portfolio	Annual Return	Volatility	Sharpe	Max Drawdown
Unconstrained Merton	35.96%	57.76%	0.59	-63.26%
Equal Weight	0.62%	12.41%	-0.11	-26.36%
Normalized Merton	12.16%	19.54%	0.52	-28.73%

Interpretation of Backtest Results

Unconstrained Allocation. The unconstrained allocation generated the highest return (35.96%) and a strong Sharpe ratio (0.59). However, the strategy relied heavily on leverage, with total portfolio weights summing to approximately 2.96. This implies significant borrowing to finance large positions across assets. While leverage amplified returns during favorable periods, it also produced substantial volatility (57.76%) and deep drawdowns (-63.26%).

Equal-Weight Benchmark. The equal-weight portfolio serves as a simple diversification benchmark requiring no parameter estimation. It exhibited significantly lower volatility (12.41%) and smaller drawdowns (-26.36%), but delivered weak returns during the test period. This reflects the broad performance of the diversified ETF basket over the evaluation window rather than any optimization effect.

Normalized Allocation. To isolate the effect of leverage, the unconstrained weights were normalized so that portfolio weights sum to one. This substantially reduced volatility and drawdowns, bringing risk levels closer to those of the equal-weight portfolio. The normalized strategy still achieved a considerably higher return (12.16%) and Sharpe ratio (0.52) than the

benchmark, suggesting that the allocation structure retains some predictive value even when leverage is removed.

Key Insights

The backtest highlights several important insights:

1. **Leverage plays a central role in theoretical optimal allocations.** The unconstrained allocation delivers strong returns primarily because the model allows leveraged positions.
2. **Optimal portfolios are highly sensitive to estimated parameters.** In practice, the parameters $\hat{\mu}$ and $\hat{\Sigma}$ must be estimated from historical data. The allocation rule

$$w^* = \frac{1}{\gamma}\hat{\Sigma}^{-1}(\hat{\mu} - r\mathbf{1})$$

   can produce extreme weights when expected return estimates are noisy.
3. **Risk-adjusted performance improves once leverage is constrained.** Normalizing weights reduces instability while retaining much of the Sharpe performance of the unconstrained allocation.
4. **Implementation bridges theory and empirical portfolio construction.** While the continuous-time model provides a clean analytical benchmark, empirical implementation requires addressing estimation error, leverage constraints, and parameter instability.

These observations motivate several practical extensions, including introducing covariance shrinkage estimators for stable weights, and long-only or leverage constraints.

Overall, the empirical results illustrate how theoretical optimal allocations interact with real-world data and highlight the importance of robust estimation techniques in practical portfolio construction.

7 Current progress and Future Directions

7.1 Covariance Shrinkage Implementation (Ongoing Work)

One of the key challenges that I observed in the empirical backtest was the sensitivity of optimal portfolio weights to estimation error in the covariance matrix. When the covariance matrix $\hat{\Sigma}$ is estimated directly from historical data, even small sampling errors led to unstable inverse covariance matrices and extreme portfolio weights. To address this issue, we can implement covariance shrinkage methods, which are known to improve the stability of covariance estimation by combining the sample covariance matrix with a structured target matrix

Let $\hat{\Sigma}$ denote the sample covariance matrix estimated from historical returns, and let F denote a structured target matrix (typically a scaled identity matrix or constant correlation matrix). The shrinkage estimator is defined as

$$\Sigma_{shrink} = (1 - \lambda)\hat{\Sigma} + \lambda F$$

where $0 \leq \lambda \leq 1$ is the shrinkage intensity parameter.

The idea is to partially 'shrink' the noisy sample covariance matrix toward a more stable target structure. When $\lambda = 0$, the estimator reduces to the original sample covariance matrix, while $\lambda = 1$ fully replaces it with the target matrix.

In practice, the shrinkage intensity can be estimated using the Ledoit–Wolf procedure, which chooses λ to minimize the mean squared error between the estimated covariance matrix and the true covariance matrix.

The new shrunk covariance matrix Σ_{shrink} is then used in the portfolio allocation formula

$$w^* = \frac{1}{\gamma}\Sigma_{shrink}^{-1}(\hat{\mu} - r\mathbf{1})$$

This approach stabilizes the inverse covariance matrix and directly addresses the instability that we observed in empirical results.

7.2 Extension to the Full Dynamic Merton Problem

The current empirical implementation focuses on terminal wealth utility under constant portfolio weights. The natural extension is the full consumption-investment problem originally formulated by Merton in 1969.

In this setting, our investor chooses both consumption c_t and portfolio allocation π_t dynamically to maximize:

$$\mathbb{E}\left[\int_0^T e^{-\rho t}\frac{c_t^{1-\gamma}}{1-\gamma}\,dt + e^{-\rho T}\frac{W_T^{1-\gamma}}{1-\gamma}\right].$$

The solution involves solving the Hamilton-Jacobi-Bellman (HJB) PDE factoring in wealth consumption and continuous-allocation. This would allow for dynamic feedback control, and much better practical modeling in real-world context. These directions align perfectly with modern quantitative asset management.